

Sepehr Ebrahimi

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Education

- BS Amirkabir University of Technology**, Electrical Engineering Oct 2022-
- Major: Telecommunication
 - GPA: 18.19/20.0 out of 105 credits
- High School Sampad (Hashemi Nejad No.4)**, Mashhad, Iran Oct 2019- Jun 2022
- Major: Mathematics and Physics
 - GPA: 19.62/20.0

Honors

- Top 10% in Electrical Engineering, among 124 other students, Amirkabir University of Technology, Tehran, Iran
- Ranked 3rd in Communication group, among 26 other students
- Ranked 503rd in university entrance exam, among more than 145,000 participants [summer 2022]

Selected Courses

- 🎧 Probability & Statistics (17/20)
- 🎧 Electromagnetics (17.75/20)
- 🎧 Signal & System (17.8/20)
- 🎧 Logic Circuits (19.5/20)
- 🎧 Electronics (II) (19/20)
- 🎧 Linear Control Systems (17/20)
- 🎧 Communication Systems (I) (17.34/20)
- 🎧 Digital Control Systems (16.5/20)
- 🎧 Digital Signal Processing (14.5/20)
- 🎧 Fields & Waves (18/20)
- 🎧 Communication Systems (II) (in progress)
- 🎧 Transmission Systems (in progress)
- 🎧 Information Theory (Graduate) (in progress)
- 🎧 Linear Algebra (self studied)
- 🎧 Machine Learning (self studied)
- 🎧 Graph Signal Processing (self studied)
- 🎧 Optimization (self studied)

Selected Projects

AM & FM Simulation

Supervisor: Prof. Hamzeh Beyranvand

- Generated an audio signal and extracted features in MATLAB; implemented AM and FM modulation/demodulation.
- Compared spectra with FFT; analyzed modulation index, frequency sensitivity, and reconstructed-signal quality.

Controller Design & Simulation for a Satellite Control-Moment Gyroscope (CMG) System

Supervisor: Prof. Heidar Ali Talebi

- Linearized the dynamics of a clustered CMG setup.
- Designed and applied controllers (including a nonlinear controller) and validated performance via simulation.

Maze Tracking

Supervisor: Prof. Heidar Ali Talebi

- Develop an E-puck robot controlled by a PID controller in Webots software
- Applied frontier based and A* algorithms for navigation

PDE Wave Equation Simulation (MATLAB)

Supervisor: Prof. Yaser Norouzi

- Solved 1-D and 2-D wave equations numerically using the Finite Difference Method.
- Implemented Dirichlet and Neumann boundary conditions; visualized time evolution and animated results.

Digital System Design in VHDL

Supervisor: Prof. Zahra Sadat Shariatmadar

- Wrote synthesizable VHDL for core modules: ALU, simple RAM, and a finite-state-machine Sequence Detector.
- Built testbenches, verified behavior in ModelSim, and prepared the design for FPGA synthesis.

Computer Skills

Programming & Scripting

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|-----------------|--------------|----------------|
| • Python | – Tensorflow | • C/C++ |
| – Pytorch | – Pandas | • TEX |
| | – Numpy | |

tools & IDEs

- | | |
|-----------------|---------------|
| • Matlab | • HFSS |
| • ADS | • VHDL |

Teaching Experiences

- | | |
|---|---|
| •  Engineering Mathematics | •  Communication Systems (I) |
| – Instructor: Prof. Yaser Norouzi | – Instructor: Prof. Yaser Norouzi |